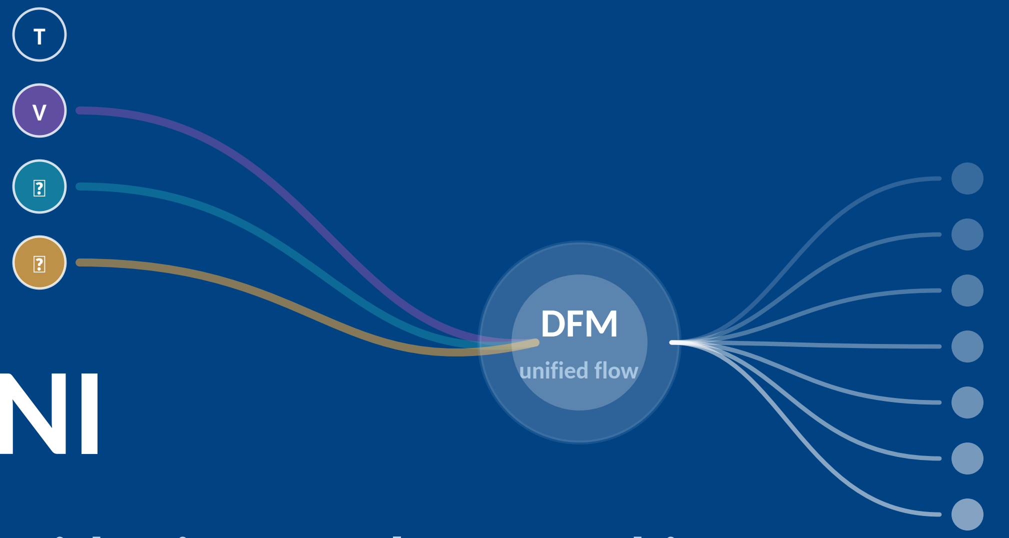


NExT-OMNI

Any-to-Any Omnimodal Foundation Models with Discrete Flow Matching

any input → any output

Run Luo · Xiaobo Xia · Lu Wang · Longze Chen · Renke Shan · Jing Luo · Min Yang · Tat-Seng Chua

SIAT, Chinese Academy of Sciences · UCAS · NExT++ Research Center · NUS

arXiv:2510.13721v2 [cs.CL]

The conflict — one model, three jobs, four modalities

01 CAUSAL BOTTLENECK

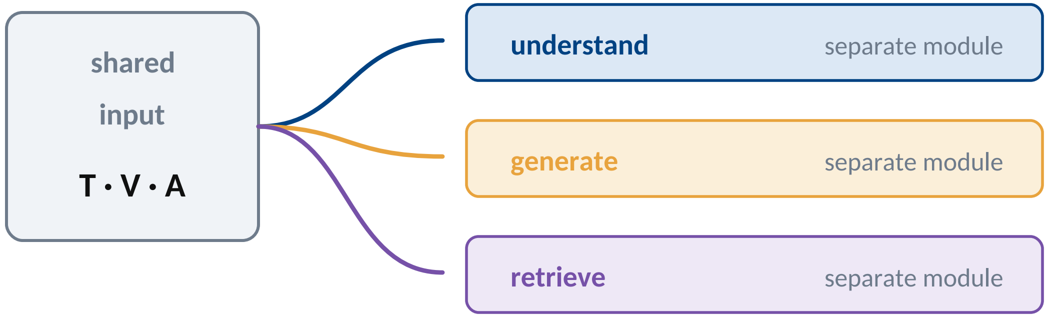
Autoregression forces every modality through one direction

token 1 must wait for token 0

understanding wants both directions

02 DECOUPLING TAX

Hybrid systems solve the conflict by duplicating pathways



features stay separated

THE TARGET

one representation for

TEXT IMAGE VIDEO AUDIO

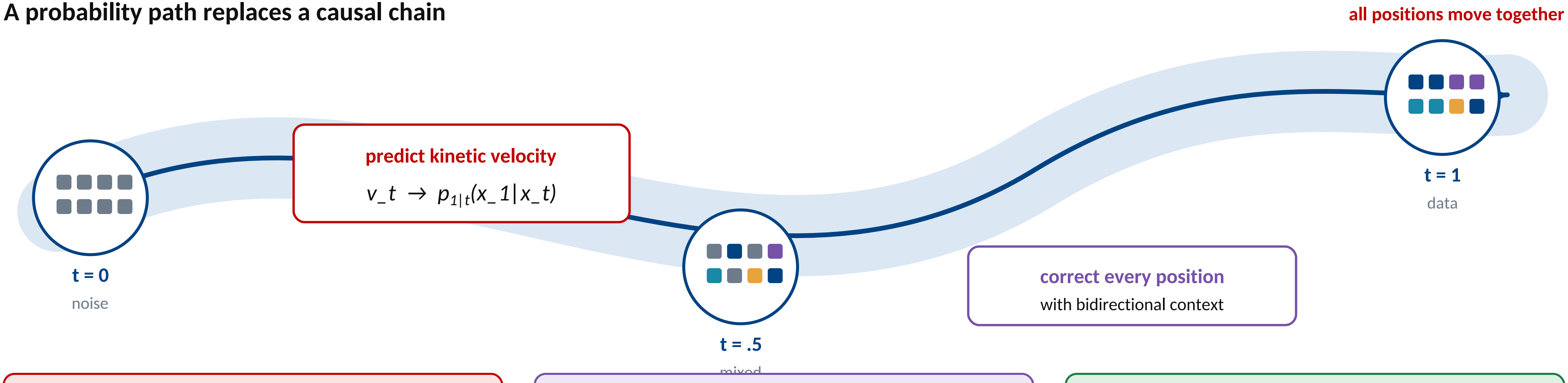
UNDERSTAND GENERATE RETRIEVE

Autoregression is sequential; hybrid systems unify tasks by splitting the model.

Introduction and Figure 1 of the paper.

The alternative — discrete flow learns to correct in parallel

A probability path replaces a causal chain

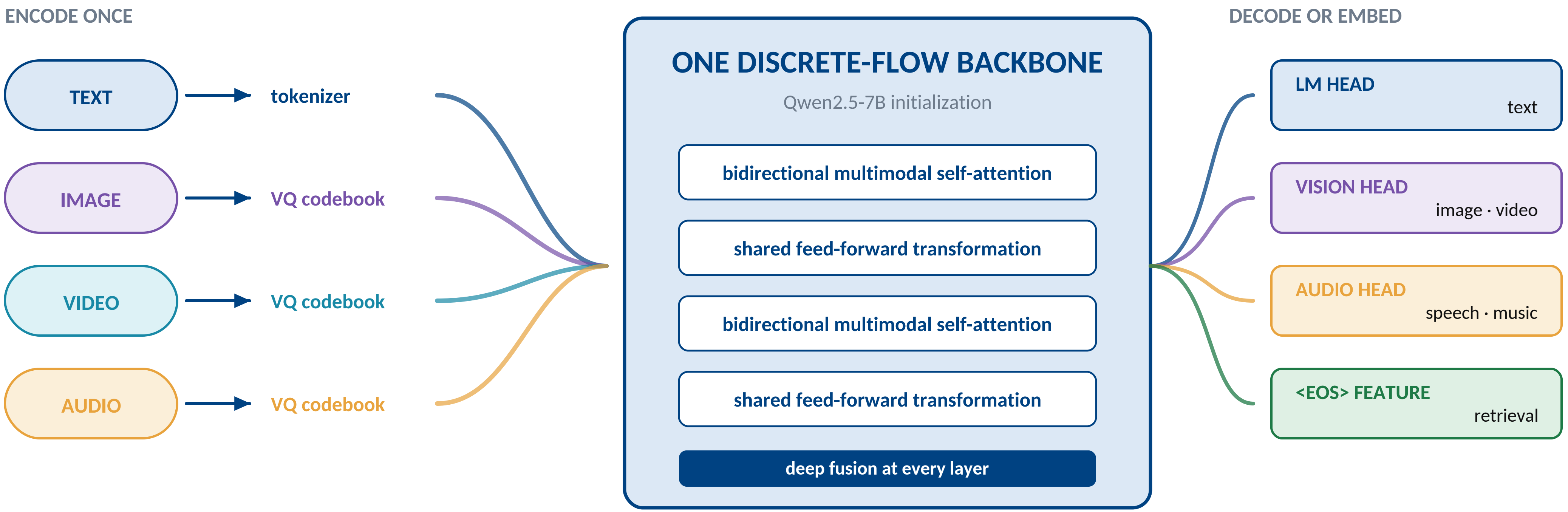


01 CORRUPT sample x▯ on a metric-induced path	**02 PREDICT** recover clean tokens at every position	**03 REFINE** iterate in parallel until t = 1

Corrupt globally, predict globally, refine iteratively — with bidirectional context.

Sections 2.3–2.4 and Appendix A of the paper.

NExT-OMNI — a single deeply fused representation



warmup $L_{rec} + L_{sem}$ joint flow training $\lambda_1 L_{CE} + \lambda_2 L^V_{rec} + \lambda_3 L^A_{rec}$ **GradNorm balances the objectives**

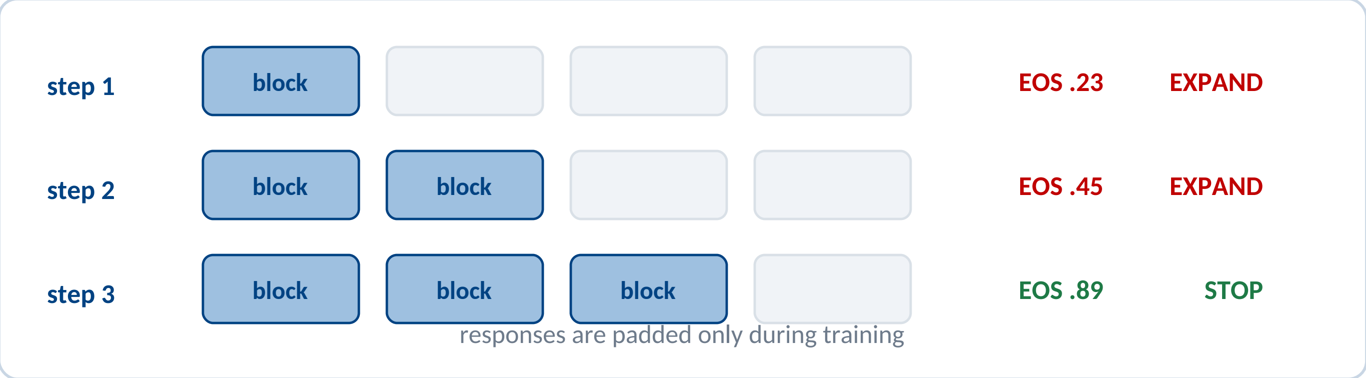
One backbone serves understanding, generation, and retrieval — no task router required.

Sections 2.1–2.3 and Figure 2 of the paper.

Training and inference — make iterative decoding practical

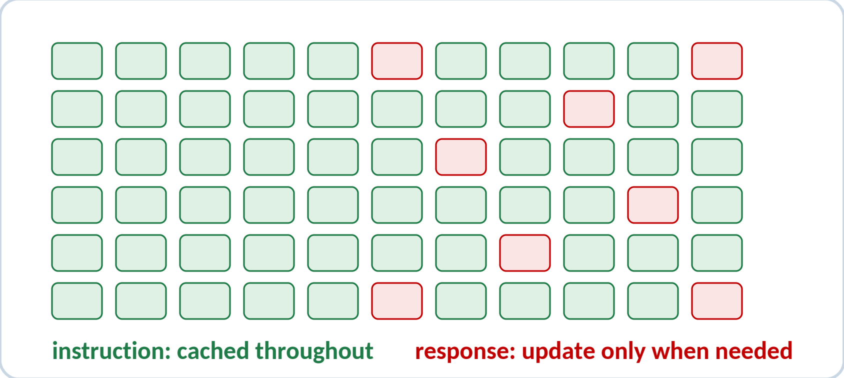
DYNAMIC RESPONSE

Generate only as many blocks as the answer needs



ADAPTIVE CACHE

Reuse features that barely move



1.4× training efficiency
single-modality batches

1.2× response speed vs AR
parallel decode + cache

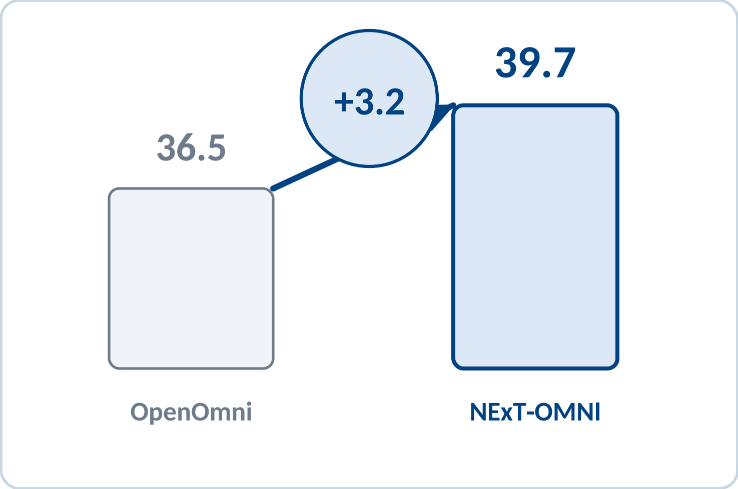
64 token block size
dynamic expansion

Interleave modalities, grow responses by blocks, and cache features that barely change.

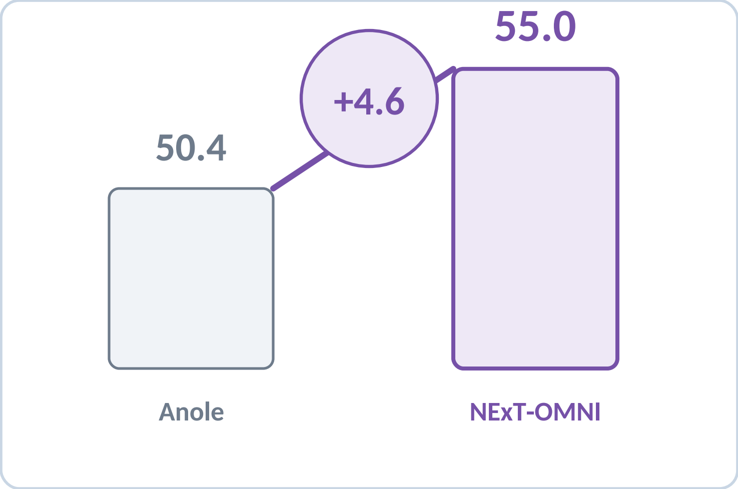
Section 2.4 and Figure 3 of the paper.

Evidence — the unified flow transfers across task families

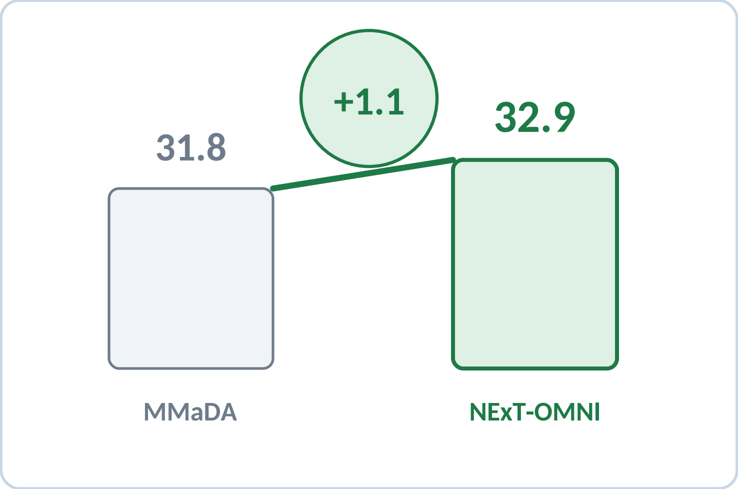
OMNIMODAL UNDERSTANDING Table 1



MULTI-TURN VISION Table 2



CROSS-MODAL RETRIEVAL Table 4



One representation transfers beyond generation

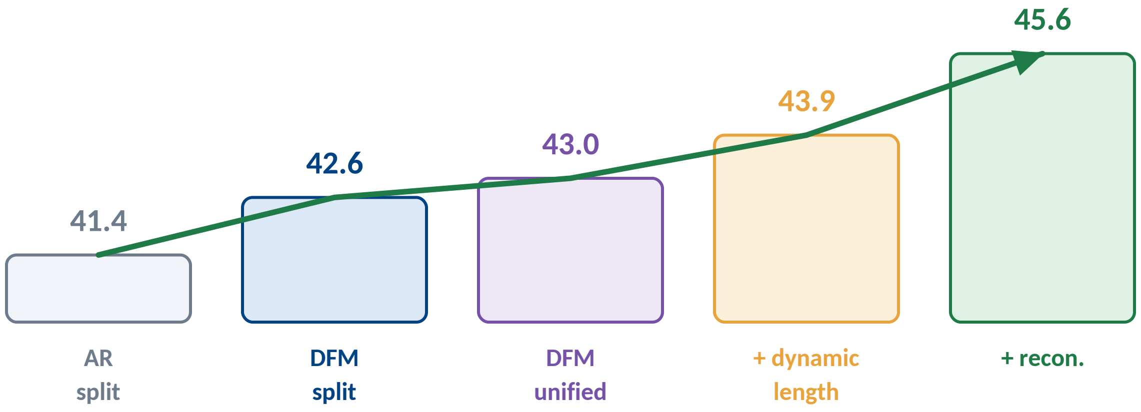
bidirectional correction + deep fusion benefit every evaluated task family

NExT-OMNI leads on understanding, multi-turn vision, and cross-modal retrieval.

Main-paper Tables 1, 2, and 4.

What the ablation says — unification needs the full recipe

THE ABLATION BUILDS A CAUSAL STORY



+4.2 average points

THE FINAL RECIPE

1. **Discrete flow** — correct globally
2. **Unified representation** — fuse deeply
3. **Dynamic length** — answer naturally
4. **Reconstruction** — keep fine detail

NEXT action trajectories · physical-AI video

SCOPE paper reports benchmark breadth, not deployed safety

DFM opens the door; dynamic length and reconstruction make the unified model work.

Run Luo, Xiaobo Xia, Lu Wang, Longze Chen, Renke Shan, Jing Luo, Min Yang, and Tat-Seng Chua. NExT-OMNI: Towards Any-to-Any Omnimodal Foundation Models with Discrete Flow Matching. arXiv:2510.13721v2, 2025.